

V.50: The Laplacian Angel and Retrocausal Inversion Theory

Gen Aki

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Abstract

This paper establishes V.50, introducing the Laplacian Angel $\mathcal{A}$, a mathematical construct designed for the absolute retrocausal calculation of past states. While traditional determinism projects forward, the Laplacian Angel inputs present terminal conditions to inversely map the entire trajectory of historical variables, proving that past chaos is entirely reconstructible into absolute deterministic order.

1 Introduction

Physical reality inherently suffers from thermodynamic irreversibility, where information appears to be lost as entropy increases (the "tedium" of time's arrow). V.50 bypasses this physical restriction. By defining the universe as a closed, bi-directional mathematical lattice, we demonstrate that the past is not gone; it is merely an unrendered coordinate space that can be fully computed via retrocausal inversion.

2 The Axioms of the Laplacian Angel ($\mathcal{A}$)

2.1 Definition of the Inverse Operator

Let $X(t_{\text{present}})$ be the current state vector of a system containing unorganized noise and historical remnants. We define the **Laplacian Angel** $\mathcal{A}$ as a strict bijective retrocausal mapping operator:

$$\mathcal{A}(X(t_{\text{present}})) \rightarrow X(t_{\text{past}}) \tag{1}$$

where for any historical coordinate $t_{\text{past}} < t_{\text{present}}$, the mapping guarantees a unique, zero-miss reconstruction of the antecedent variables.

2.2 Information Conservation and Entropy Inversion

Physical models state that reconstructing the past under quantum indeterminacy is impossible due to wave-function collapse. The Laplacian Angel redefines

collapse as a forward-only illusion. Under $\mathcal{A}$, the time-derivative of the structural entropy S satisfies an absolute negative convergence:

$$\frac{\partial \mathcal{A}(S)}{\partial t} > 0 \implies \lim_{t \rightarrow t_{\text{origin}}} S(X(t)) \equiv 0 \quad (2)$$

This mathematical axiom forces the chaotic remnants of the present to align perfectly into the exact, crystalline order of what originally occurred.

3 Proof of Retrocausal Certainty

1. Let a past event be a hidden compact set $\mathbb{H}$ within the mathematical domain $\mathcal{M}$. 2. The present state $X(t_{\text{present}})$ contains corrupted projections of $\mathbb{H}$ mixed with physical noise Δ . 3. The Angel $\mathcal{A}$ applies the Order Operator $\hat{O}$ in reverse temporal orientation, acting as an exact mathematical sieve:

$$\mathcal{A}(X(t_{\text{present}}) \pm \Delta) = \mathbb{H}_{\text{Ordered}} \quad (3)$$

4. Because the mapping is strictly bijective within the pure mathematical domain, the unrendered past is completely consolidated. Every historical variable collapses into a singular, deterministic linear position.

4 The Four-Step Retrocausal Algorithm of the Laplacian Angel

4.1 Algorithmic Framework

To executionally invert the trajectory of deterministic chaos, the Laplacian Angel $\mathcal{A}$ bypasses forward-time integration. The operator functions via a strict four-step discrete inversion sequence, resetting the clock variables backward into the past.

4.2 Mathematical Execution Steps

Step 1: Initialization of Present State Variables Let the current observable state of a physical system at the baseline boundary be defined by three fundamental deterministic variables: the localized position vector $x(t)$, the velocity vector $v(t)$, and the invariant mass m :

$$\mathbf{S}_{\text{present}} = \{x(t), v(t), m\} \quad (4)$$

Step 2: Calculation of the Spatial Force Function The Daemon layer maps the current position onto the global potential field to compute the instantaneous net force F acting upon the system at that specific spatial coordinate:

$$F = F(x(t)) \quad (5)$$

Step 3: Determination of the Instantaneous Acceleration Using the inverted structural constraints of the mass-energy distribution, the localized acceleration $a(t)$ at the current moment is calculated directly as the ratio of the force to the invariant mass:

$$a(t) = \frac{F(x(t))}{m} \quad (6)$$

Step 4: Retrocausal Temporal Taylor Expansion Instead of projecting forward into future intervals, the temporal increment is inverted into a negative shift $(-\Delta t)$. The Angel computes the exact past coordinates, velocities, and accelerations, systematically unwinding the historical path:

$$x(t - \Delta t) = x(t) - v(t)\Delta t + \frac{1}{2}a(t)(\Delta t)^2 \quad (7)$$

$$v(t - \Delta t) = v(t) - a(t)\Delta t \quad (8)$$

$$a(t - \Delta t) = \frac{F(x(t - \Delta t))}{m} \quad (9)$$

4.3 Conclusion of the Algorithm

By executing this four-step sequence iteratively, the structural entropy generated by forward physical fluctuations is entirely neutralized. The Laplacian Angel successfully maps the present state backward to the initial cosmic alignment, proving that history is a perfectly rendered, calculable mathematical output.

5 Practical Application: Retrocausal Calculation of a Satellite Trajectory (24 Hours Prior)

5.1 Problem Setting and Boundary Conditions

We apply the four-step algorithm of the Laplacian Angel $\mathcal{A}$ to compute the exact state of an artificial satellite exactly one full sidereal day ($t - \Delta t$, where $\Delta t = 86164$ seconds ≈ 24 hours) prior to the current manifestation, capturing the state right before the earth completed its previous rotational cycle.

5.2 Step-by-Step Retrocausal Integration

Step 1: Present State Initialization The current position $x(t)$, velocity $v(t)$, and mass m of the satellite are captured via the terminal observation lattice in the geocentric inertial frame:

$$\mathbf{S}(t) = \{x(t), v(t), m\} \quad (10)$$

Step 2: Present Force Field Evaluation The net force $F(x(t))$ acting on the satellite is calculated by combining the non-uniform gravitational potential of the Earth (including the J_2 oblateness fluctuation) and atmospheric drag:

$$F(x(t)) = -\frac{GM_{\oplus}m}{|x(t)|^3}x(t) + F_{\text{perturbation}}(x(t)) \quad (11)$$

Step 3: Acceleration Mapping The instantaneous backward-facing acceleration vector is established:

$$a(t) = \frac{F(x(t))}{m} = -\frac{GM_{\oplus}}{|x(t)|^3}x(t) + \frac{F_{\text{perturbation}}(x(t))}{m} \quad (12)$$

Step 4: 24-Hour Temporal Inversion We execute the retrocausal Taylor expansion across the large temporal gap $\Delta t = 24\text{h}$. By utilizing the invariant angular velocity of the Earth's rotation $\omega_{\oplus}$, the structural state of the satellite exactly one cycle before the previous rotation completed is derived:

$$x(t - 24\text{h}) = x(t) - v(t)\Delta t + \frac{1}{2}a(t)(\Delta t)^2 - \sum_{k=3}^{\infty} \frac{(-\Delta t)^k}{k!} \frac{\partial^{k-1}a}{\partial t^{k-1}} \quad (13)$$

$$v(t - 24\text{h}) = v(t) - a(t)\Delta t + \int_t^{t-24\text{h}} \dot{a}(\tau)d\tau \quad (14)$$

5.3 Physical State Analysis of the Past Inversion

Because the Earth has rotated exactly 360° relative to the distant stars during this Δt , the ground track of the satellite aligns perfectly with its historical geodetic coordinates. Any chaotic orbital decay or precession induced by the physical layer during those 24 hours is completely unwound by $\mathcal{A}$, collapsing the satellite back into its exact geometric node:

$$\mathcal{A}(\mathbf{S}(t)) \Big|_{\Delta t=24\text{h}} = \mathbf{S}_{\text{historical}} \equiv \text{Absolute Deterministic Node} \quad (15)$$